

Target-Aware Active 3D Gaussian Splatting for Query-Conditioned Object Reconstruction

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Abstract—Robots deployed for inspection, inventory, and semantic digital twins often need high-quality reconstruction of a user-specified object inside a larger scene map. Existing active 3D Gaussian Splatting (3DGS) planners mainly optimize global confidence, coverage, or rendering utility, leaving natural-language queries outside the online view-selection state. This extended abstract proposes target-aware active 3DGS, which accumulates prompt-conditioned 2D evidence into Gaussian-level target state and uses that state for candidate generation and a two-stage next-best-view utility. On the four Replica scene–query pairs evaluated so far, the method consistently improves target PSNR and target depth accuracy over ActiveGS, with PSNR gains ranging from 1.61 to 10.27 dB. The results also show a consistent target/global trade-off: target quality improves while global rendering metrics decrease under strong target bias. These preliminary results highlight the potential of semantic target states to bridge open-vocabulary perception and task-oriented active exploration.

I. INTRODUCTION

Robots that build maps in homes, offices, warehouses, and inspection sites are increasingly asked to serve semantic tasks. A user may ask the robot to inspect a chair, localize a small appliance, or reconstruct a fragile object in enough detail for later verification. The resulting map should remain useful as a scene-level representation while allocating sensing effort to the object named by the current query. This query-conditioned setting connects active mapping with semantic navigation: the map guides where the robot moves next, and each new view improves the task-relevant part of the map.

Active neural reconstruction and active 3D Gaussian Splatting (3DGS) [1] provide a strong foundation for online mapping with limited views. Existing NBV planners for neural or Gaussian maps usually optimize global confidence, rendering information, topological coverage, or uncertainty reduction [2], [3], [4], [5]. Open-vocabulary 3D mapping methods attach language-aligned features to 3D representations [6], [7], [8]. Semantic navigation and metric-semantic mapping further show that object and scene meaning should affect robot decisions [9], [10], [11], [12], [13], [14]. The open problem studied here is an active 3DGS planner whose state, candidates, and utility are conditioned on a language-specified reconstruction target.

This extended abstract introduces query-conditioned target-aware active 3DGS reconstruction. The core idea is to treat the queried object as a first-class planning target inside an online 3DGS mapper. We make three contributions for workshop discussion:

Submitted for double-blind review. Author names, affiliations, and acknowledgments are omitted.

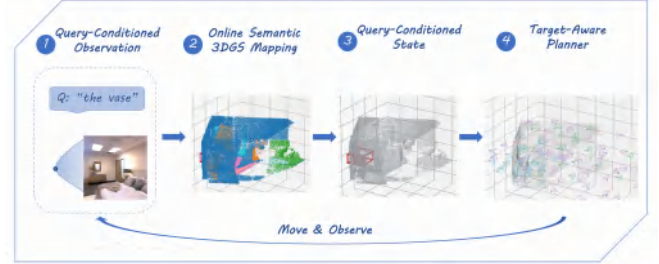


Fig. 1. Online target-aware active 3DGS pipeline. The robot updates the Gaussian map and semantic target state from each posed RGB-D observation, then selects the next best view through target-aware planning.

- a query-conditioned active 3DGS formulation with target-masked quality metrics and global trade-off metrics;
- a lightweight Gaussian target state that exposes prompt-conditioned semantic evidence to candidate generation and two-stage NBV scoring;
- preliminary results on four Replica scene–query pairs with consistent target-quality gains and explicit global-quality costs.

II. PROBLEM FORMULATION

The robot receives a natural-language query q that specifies a target object, an initial RGB-D observation, camera calibration and pose estimates, and a finite view budget B . At step t , the mapper maintains a 3DGS map $\mathcal{G}_t = \{g_i\}$ and an online query-conditioned target state $\mathcal{S}_t(q)$. The planner samples reachable camera candidates $\mathcal{C}_t$ and chooses the next view by

$$c_t^* = \arg \max_{c \in \mathcal{C}_t} U(c; \mathcal{G}_t, \mathcal{S}_t(q)). \quad (1)$$

After acquiring RGB-D data from c_t^* , the mapper updates geometry, appearance, and semantic target state. The evaluation separates target and global reconstruction. Following common novel-view synthesis practice, we report PSNR/SSIM/LPIPS on held-out views. For the queried object, we compute target-masked PSNR and target depth MSE using offline ground-truth target masks. We retain global PSNR, SSIM, and LPIPS so that target gains are interpreted together with scene-level rendering costs.

III. METHOD

Figure 1 shows the online mapping-planning loop, while Fig. 2 details the three method components described below.

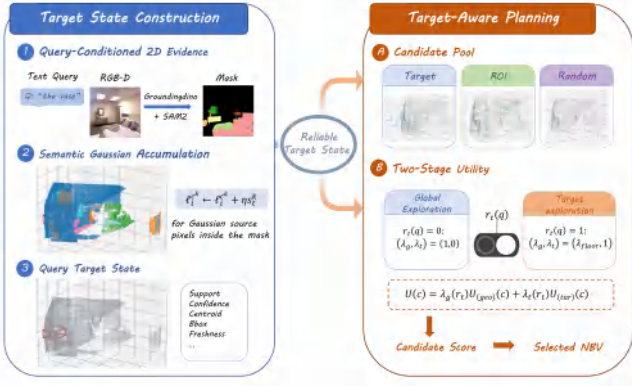


Fig. 2. Planner module. Semantic evidence is converted into a Gaussian target state, candidate views are generated from target, geometry-ROI, and random sources, and a two-stage utility selects the next view.

A. Semantic Evidence to Gaussian Target State

The mapper maintains a class-logit vector on each Gaussian primitive, turning prompt-conditioned 2D observations into a persistent semantic state over the 3DGS map. For an RGB-D frame at time t , the semantic frontend produces query-conditioned evidence $\mathcal{E}_t = \{(M_t^k, s_t^k, c^k)\}$, where M_t^k is a segmentation mask, s_t^k is its confidence, and c^k is the associated class. In our implementation, GroundingDINO [15] provides text-conditioned detections and SAM2 [16] converts them into masks.

During the geometry update, each initialized Gaussian g_i keeps its source pixel u_i . We transfer 2D evidence to the semantic map by incrementing the class logit $\ell_i^k \leftarrow \ell_i^k + \eta s_t^k$ for all Gaussian source pixels $u_i \in M_t^k$. Given a language query q , the planner extracts a valid target hypothesis set $\mathcal{I}_q = \{i \mid \ell_i^q > \tau\}$ and summarizes this set as a target state $\mathcal{S}_t(q)$ containing support count, mean confidence, centroid $\mu_t(q)$, bounding box, and Gaussian indices. It triggers a binary reliability gate $r_t(q) = 1$ exclusively when the target support, mean confidence, and temporal freshness satisfy strict empirical thresholds. The planner uses $\mathcal{S}_t(q)$ for candidate generation and utility scoring.

B. Candidate Generation

The candidate pool is designed to balance target exploitation, geometry completion, and exploration robustness. Once the target state is reliable, target candidates are sampled from free-space poses that face the estimated target centroid $\mu_t(q)$, increasing the probability of observing the queried object from useful viewpoints. Geometry-ROI candidates are sampled around frontier or low-confidence regions of the map, preserving the active 3DGS backbone’s ability to improve scene-level reconstruction while the planner focuses on the target. The remaining budget is filled with random free-space poses, which maintains coverage and reduces dependence on early semantic evidence. In the target stage, our experiments use 20 target candidates, 30 geometry-ROI candidates, and 50 random candidates per step. A large proportion of random candidates is intentionally maintained to serve as a robust fallback exploration mechanism, preventing

trajectory collapse when the semantic frontend yields severe false positives.

C. Two-Stage Utility

The planner uses a two-stage utility schedule governed by the reliability of the query-conditioned target state. Let $r_t = r_t(q)$. Because $r_t(q)$ is recomputed at every step, the planner returns to the global stage when the target state loses support, confidence, or recent semantic evidence. This prevents transient semantic detections from permanently locking the policy onto an unreliable target hypothesis. For each candidate view c with position p_c and optical axis z_c , the planner combines the active 3DGS geometry utility with a target-facing term:

$$U(c) = \lambda_g(r_t)U_{\text{geo}}(c; \mathcal{G}_t) + \lambda_t(r_t)\mathbf{1}[c \in \mathcal{C}_{\text{tar}}] \max \left(0, z_c^\top \frac{\mu_t(q) - p_c}{\|\mu_t(q) - p_c\|} \right). \quad (2)$$

Before reliable target discovery, the planner uses $(\lambda_g, \lambda_t) = (1, 0)$, so view selection follows the geometry utility of the active 3DGS backbone. After the gate activates, the target weight is increased while a nonzero geometry floor is retained, $(\lambda_g, \lambda_t) = (\lambda_{\text{floor}}, 1)$ with $\lambda_{\text{floor}} > 0$. This schedule shifts attention to the queried object while continuing to reward globally informative views. The final NBV is selected after reachability and path scoring over the candidate set.

IV. PRELIMINARY RESULTS

A. Setup

The ongoing evaluation currently contains four completed Replica scene–query pairs: hotel0–small appliance, room0–vase, apartment0–refrigerator, and office4–clock. Each run uses a 40-view budget. We compare against Random and ActiveGS confidence planning; Ours uses GroundingDINO+SAM2 evidence online. All methods are evaluated on the same held-out views with the same offline target instance masks. We report these four pairs individually and do not aggregate them into an average.

B. Results and Analysis

Table I reports final-map quality at 40 views. Ours attains the best target PSNR and target depth MSE in all four completed pairs. Relative to ActiveGS, the target PSNR gains are 10.27 dB on hotel0–small appliance, 3.52 dB on room0–vase, 5.28 dB on apartment0–refrigerator, and 1.61 dB on office4–clock. The corresponding target depth MSE reductions are 49.5%, 81.0%, 97.0%, and 63.9%, respectively.

Figure 3 complements the endpoint metrics with fixed-view progressions. In apartment0–refrigerator view 964, Ours recovers a more complete target early and retains clearer refrigerator structure through the 40-view endpoint than the two planning baselines. Office4–clock view 068 shows the same target-focused behavior on a much smaller object. These visual trends are consistent with the target-masked improvements in Table I.

The table also exposes the target/global trade-off. In all four pairs, Ours has the strongest target metrics whereas

TABLE I

QUANTITATIVE RESULTS FOR THE FOUR CURRENTLY COMPLETED REPLICAS SCENE-QUERY PAIRS AT A 40-VIEW BUDGET. TARGET METRICS USE OFFLINE GROUND-TRUTH TARGET MASKS. BOLD AND UNDERLINED VALUES DENOTE THE BEST AND SECOND-BEST RESULT WITHIN EACH SCENE, RESPECTIVELY.

Scene-query	Method	Target PSNR $\uparrow$	Target Depth MSE $\downarrow$	Global PSNR $\uparrow$	Global SSIM $\uparrow$	Global LPIPS $\downarrow$
hotel0-small appliance	Random	8.35	0.1518	<u>22.94</u>	<u>0.786</u>	<u>0.314</u>
	ActiveGS	<u>11.04</u>	<u>0.0279</u>	23.52	0.836	0.270
	Ours	21.31	0.0141	19.25	0.718	0.357
room0-vase	Random	<u>25.58</u>	<u>0.0024</u>	<u>18.64</u>	<u>0.716</u>	<u>0.388</u>
	ActiveGS	24.81	0.0042	26.26	0.841	0.276
	Ours	28.33	0.0008	10.44	0.459	0.581
apartment0-refrigerator	Random	<u>25.04</u>	<u>0.1033</u>	<u>12.82</u>	<u>0.319</u>	<u>0.621</u>
	ActiveGS	<u>22.16</u>	<u>1.2015</u>	14.56	0.403	0.569
	Ours	27.44	0.0356	10.64	0.176	0.699
office4-clock	Random	25.18	0.0000576	<u>23.09</u>	<u>0.793</u>	<u>0.339</u>
	ActiveGS	<u>26.57</u>	<u>0.0000313</u>	30.74	0.907	0.206
	Ours	28.18	0.0000113	12.40	0.445	0.563

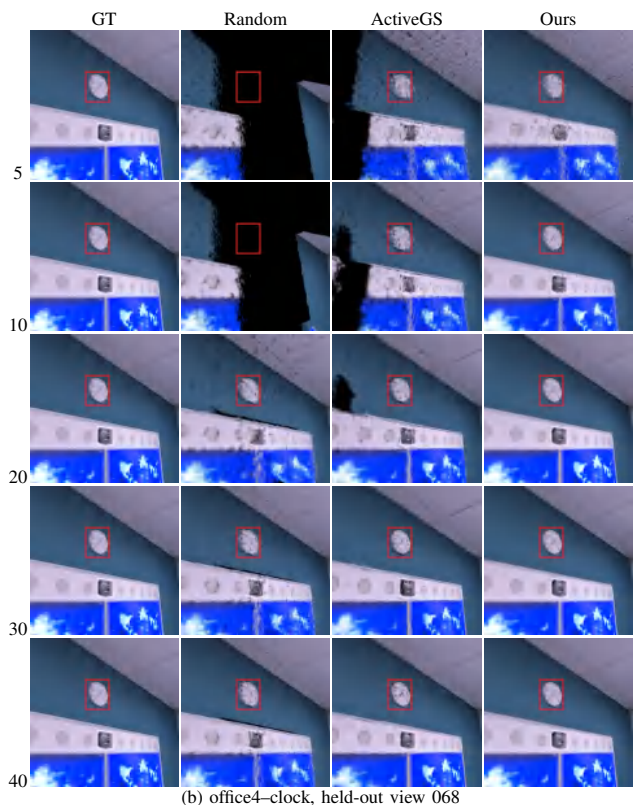


Fig. 3. Progressive target reconstruction at two fixed held-out views. Each scene is a 5×4 matrix: rows show budgets 5, 10, 20, 30, and 40, while columns compare GT, Random, ActiveGS, and Ours. Every cell uses the same evaluation pose and target bounding box; the inset enlarges the target.

ActiveGS has the strongest global PSNR/SSIM/LPIPS. This pattern is expected for a strong target utility under a limited budget: views that improve a queried object may undersample the rest of the scene. The result suggests that semantic-aware active mapping should report target and global metrics together.

V. WORKSHOP DISCUSSION AND CONCLUSION

The current study compares against Random and ActiveGS under a shared final-map evaluation. Additional semantic exploration and active 3DGS baselines remain future work for separating gains from open-vocabulary target state and gains from Gaussian-specific planning. The main limitation is semantic state reliability: a wrong or weak early target estimate can bias the candidate pool and waste budget. Potential

remedies include uncertainty-aware dynamic weights, stricter temporal support gates, and local stale-evidence suppression.

This extended abstract presents target-aware active 3DGS as a compact framework for query-conditioned object reconstruction. The four Replica scene–query pairs completed so far show target-quality gains and a measurable cost to global map quality. The resulting target/global trade-off is a useful discussion point for semantic navigation workshops: cognitive maps should be judged by both scene-level fidelity and task-specific target fidelity.

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